

Erdős #683: reduction to prime-free intervals, and identification with the Sylvester–Erdős problem

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Abstract

#683 asks whether $P\binom{n}{k} \geq \min(n - k + 1, k^{1+c})$ for some $c > 0$, where P denotes the largest prime factor. We prove that the inequality holds automatically — for every $c > 0$ — whenever $(n - k, n]$ contains a prime, so the problem reduces entirely to prime-free intervals; by Baker–Harman–Pintz this leaves only $k < n^{0.525+\varepsilon}$. We further show that in its main range #683 is the Sylvester–Erdős problem on the largest prime factor of a product of consecutive integers. #683 itself is not proved.

1 Setup

By Kummer, $v_p\binom{n}{k}$ is the number of carries in the base- p addition $k + (n - k)$; the set of primes dividing $\binom{n}{k}$ is $\mathcal{D}(n, k) = \{p : k \not\leq_p n\}$, where $k \leq_p n$ means every base- p digit of k is at most that of n . #683 asks for $\max \mathcal{D}$. Throughout $1 \leq k \leq n/2$.

2 The window

Lemma 1 (window). *Let $p > n/2$ be prime. Then $v_p\binom{n}{k} = 1$ if $n - k < p \leq n$, and $v_p\binom{n}{k} = 0$ if $n/2 < p \leq n - k$.*

Proof. For $p > n/2$ and $m \leq n$ we have $p^2 > n \geq m$, so $v_p(m!) = \lfloor m/p \rfloor$. Hence $v_p\binom{n}{k} = \lfloor n/p \rfloor - \lfloor k/p \rfloor - \lfloor (n - k)/p \rfloor$, in which $\lfloor n/p \rfloor = 1$, $\lfloor k/p \rfloor = 0$ (as $k \leq n/2 < p$), and $\lfloor (n - k)/p \rfloor$ is 1 or 0 according as $p \leq n - k$ or not. $\square$

Verified: no exception, in either direction, for $4 \leq n < 420$ and all $k \leq n/2$.

Theorem 2 (reduction). *If $(n - k, n]$ contains a prime then $P\binom{n}{k} \geq n - k + 1$, so the inequality of #683 holds for every $c > 0$. Consequently #683 is equivalent to its restriction to pairs (n, k) for which $(n - k, n]$ is prime-free.*

Proof. A prime $q \in (n - k, n]$ divides $\binom{n}{k}$ by Lemma 1, so $P\binom{n}{k} \geq q > n - k$, and the minimum is attained by its first branch or exceeded. $\square$

Measured: among the 2 248 480 pairs with $10 \leq n < 3000$, $1 \leq k \leq n/2$, the interval $(n - k, n]$ is prime-free for 13 909 — a proportion of 0.0062. So Theorem 2 settles 99.38% of the range unconditionally.

Corollary 3 (unconditional range). *Fix $\varepsilon > 0$. For n large and $n^{0.525+\varepsilon} \leq k \leq n/2$, the interval $(n-k, n]$ contains a prime by Baker–Harman–Pintz, so #683 holds there for every $c > 0$. Only $k < n^{0.525+\varepsilon}$ remains.*

Measured: the longest prime-free $(n-k, n]$ below $n = 2 \times 10^4$ has length 51, against $n^{0.525} = 181$.

3 Identification with Sylvester–Erdős

Theorem 4 (refined window). *Let $p > \sqrt{n}$ be prime with $p > k$. Then $p \mid \binom{n}{k}$ if and only if p has a multiple in $(n-k, n]$.*

Proof. $p^2 > n$ gives $v_p(m!) = \lfloor m/p \rfloor$ for $m \leq n$, and $k < p$ gives $\lfloor k/p \rfloor = 0$; so $v_p\left(\binom{n}{k}\right) = \lfloor n/p \rfloor - \lfloor (n-k)/p \rfloor$, the number of multiples of p in $(n-k, n]$. $\square$

Verified: 7 327 302 triples (n, k, p) with $n < 700$ — no mismatch.

Corollary 5. *If $P^+(\prod_{m=n-k+1}^n m) > \max(\sqrt{n}, k)$ then $P\left(\binom{n}{k}\right) = P^+(\prod_{m=n-k+1}^n m)$. Hence in its main range #683 is exactly the Sylvester–Erdős problem on the largest prime factor of a product of k consecutive integers near n .*

Verified: equality with no mismatch for $20 \leq n < 500$, all $k \leq n/2$ with the set nonempty. *Measured:* over $n < 1200$, $3 \leq k < 60$, restricted to pairs where the binding branch is k^{1+c} , the least admissible exponent is $c = 0.4650$, at $(n, k) = (10, 3)$ with $P\left(\binom{10}{3}\right) = P(120) = 5$; the extremal case is not drifting upward with n .

4 A ledger bound forcing $P \gg k \log k$

The reduction of §2 disposes of the range where $(n-k, n]$ contains a prime. On the complementary range the size of $\binom{n}{k}$ and the ledger bound each rail’s contribution, and the two together bound P from below.

Lemma 6 (rail ceiling). *For every prime p and $1 \leq k \leq n$,*

$$v_p\left(\binom{n}{k}\right) \leq \left\lfloor \frac{\log n}{\log p} \right\rfloor + 1.$$

Proof. By Kummer, $v_p\left(\binom{n}{k}\right)$ is the number of carries in the base- p addition $k + (n-k) = n$. Carries occur only out of positions $0, 1, \dots, \lfloor \log_p n \rfloor$, since n has $\lfloor \log_p n \rfloor + 1$ base- p digits and no carry leaves the top position. $\square$

Verified: 660 046 triples (n, k, p) with $n < 260$ — no violation.

Theorem 7 (the smoothness inequality). *Let $2 \leq k \leq n/2$ and put $y = P\left(\binom{n}{k}\right)$. Then*

$$\frac{k \log(n/k)}{\log y} \leq \Omega\left(\binom{n}{k}\right) \leq \log n \sum_{p \leq y} \frac{1}{\log p}.$$

Proof. For the left inequality, $\binom{n}{k} \geq (n/k)^k$, and any integer $N > 1$ all of whose prime factors are at most y satisfies $N \leq y^{\Omega(N)}$, whence $\Omega(N) \geq \log N / \log y$.

For the right inequality, every prime factor of $\binom{n}{k}$ is at most y , so by Lemma 6

$$\Omega\left(\binom{n}{k}\right) = \sum_{p \leq y} v_p\left(\binom{n}{k}\right) \leq \sum_{p \leq y} \left(\frac{\log n}{\log p} + 1\right),$$

and the term $\pi(y)$ is absorbed since $\log n / \log p \geq 1$ for every $p \leq n$. $\square$

Verified: 16 510 pairs (n, k) with $n < 260$ — both inequalities hold throughout.

Corollary 8. *For $k \leq \sqrt{n}$ one has $P\left(\binom{n}{k}\right) \gg k \log k$.*

Proof. By the prime number theorem $\sum_{p \leq y} 1/\log p \sim y/(\log y)^2$. Substituting into Theorem 7 and using $\log(n/k) \geq \frac{1}{2} \log n$ for $k \leq \sqrt{n}$ gives

$$\frac{k \log n}{2 \log y} \leq (1 + o(1)) \frac{y \log n}{(\log y)^2}, \quad \text{i.e.} \quad y \geq (1 + o(1)) \frac{1}{2} k \log y.$$

Since $y > k$ by Sylvester–Schur, $\log y > \log k$, and the claim follows. $\square$

This recovers Erdős’s bound $P\left(\binom{n}{k}\right) \gg k \log k$ from the ledger alone, with no input beyond Kummer, Legendre and the prime number theorem. The gap to #683 is the passage from $k \log k$ to k^{1+c} : Theorem 7 is sharp in the exponent of y , so any improvement must come from a better upper bound on $\Omega\left(\binom{n}{k}\right)$ than the rail ceiling of Lemma 6, which is attained only when every position carries.

Remark 9 (register). **Proved:** *Lemmas and Theorems above, all elementary consequences of Legendre and Kummer; Corollary 3 additionally cites Baker–Harman–Pintz, and Corollary 8 the prime number theorem. Lemma 6, Theorem 7 and Corollary 8 recover Erdős’s $P\left(\binom{n}{k}\right) \gg k \log k$ from the ledger alone. None is claimed to be new — Lemma 1 is surely folklore. What is recorded is the reduction to prime-free intervals and the identification in Corollary 5.* **Measured:** *the 0.62% proportion, $c = 0.4650$, the length-51 window.* **Not proved:** #683. **Falsifier:** *the least admissible c declining with the range would indicate no fixed c works; it did not.*